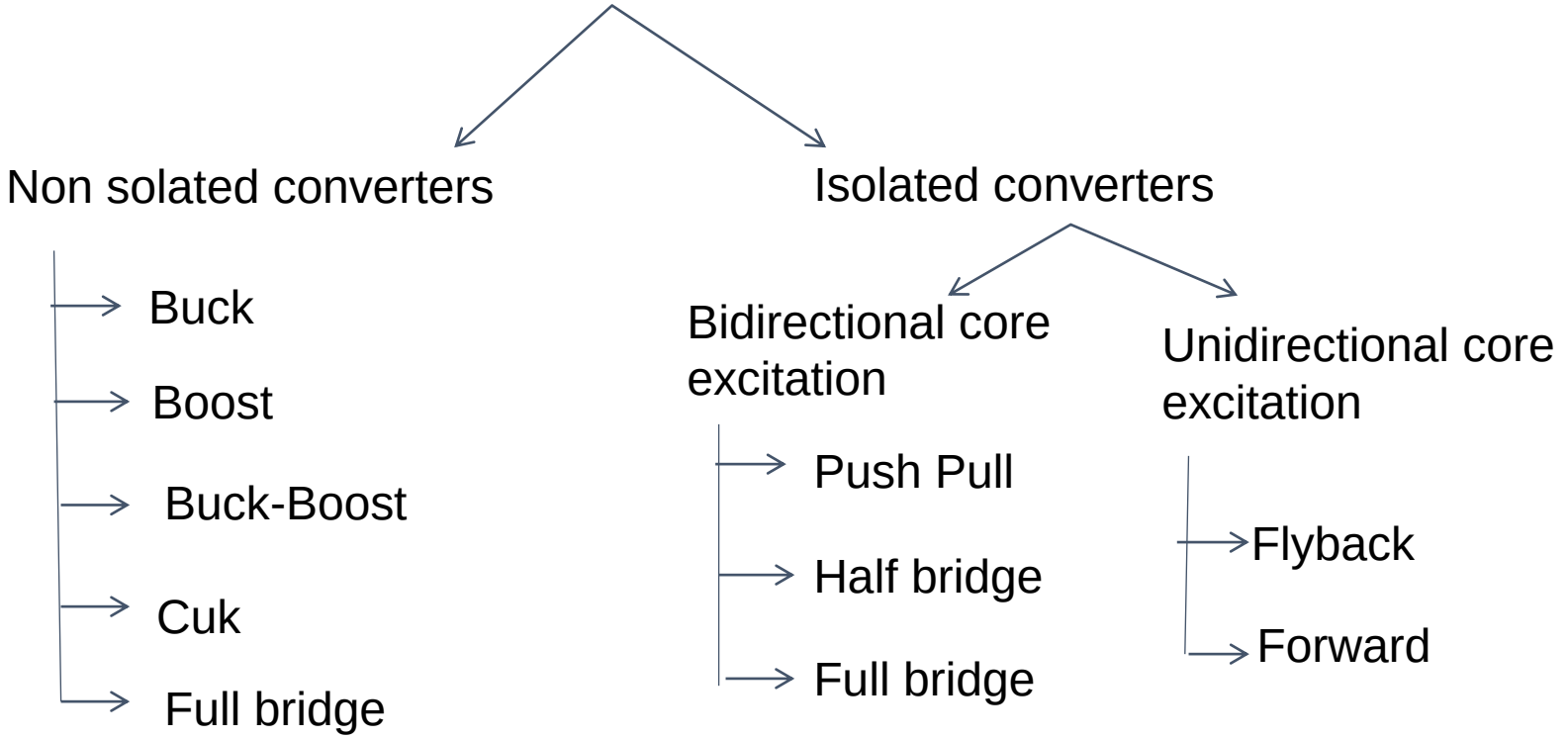
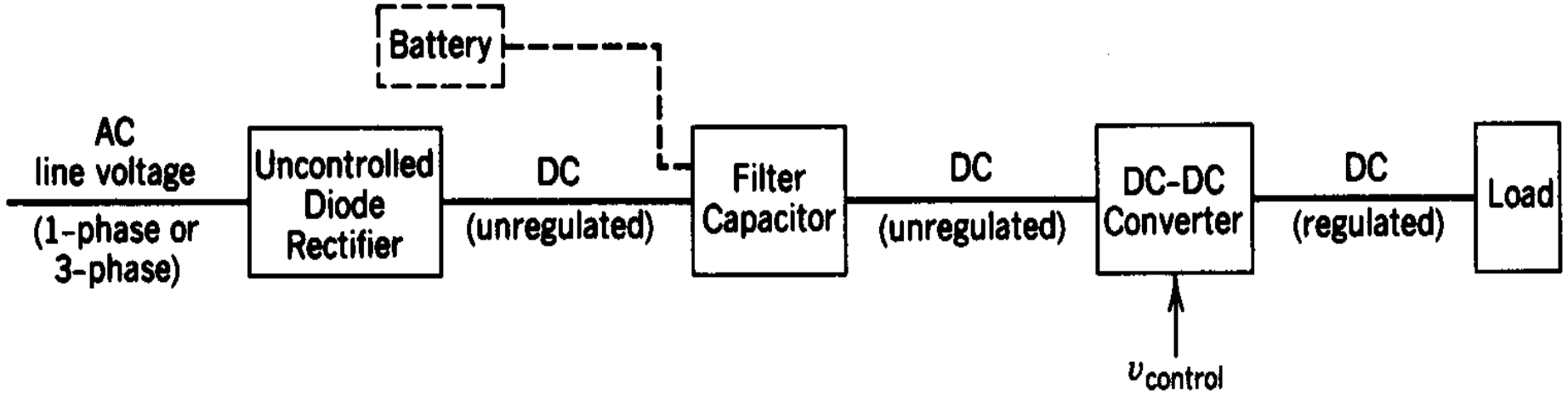
DC-DC Switch-Mode Converters

- To convert unregulated dc i/p into controlled dc o/p at a desired voltage level
- Used to regulate switch mode dc power supplies and in dc drive applications

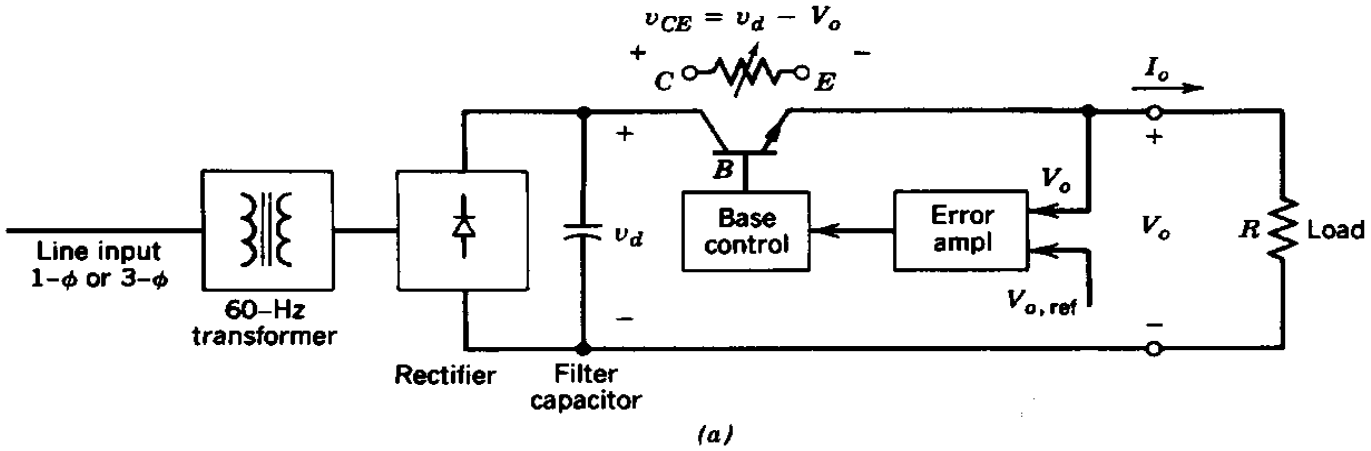
DC-DC Switch-Mode Converters



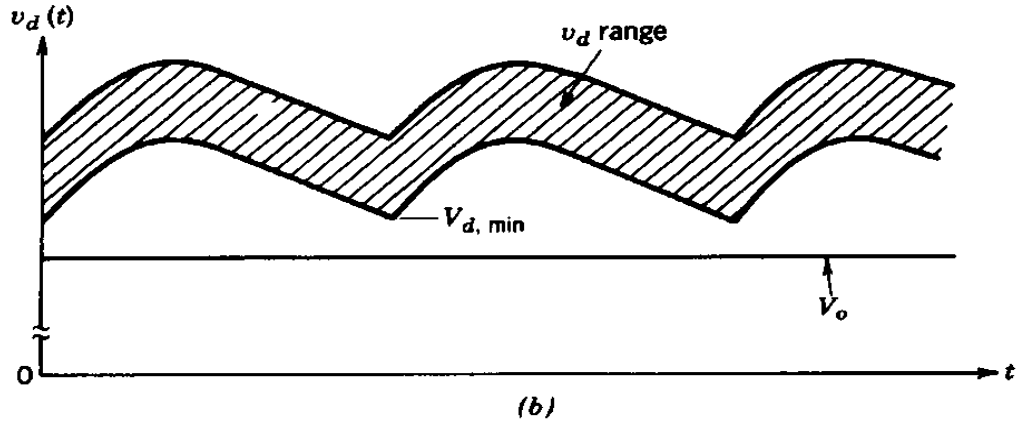
Block Diagram of DC-DC Converters



Linear Power Supplies



(a)



(b)

• Disadvantages - Very poor efficiency and large weight and size- requires low frequency transformers

•Advantages - Simple circuitary, Less cost for small power rating, Do not produce large EMI

Switching DC Power Supply: Block Diagram

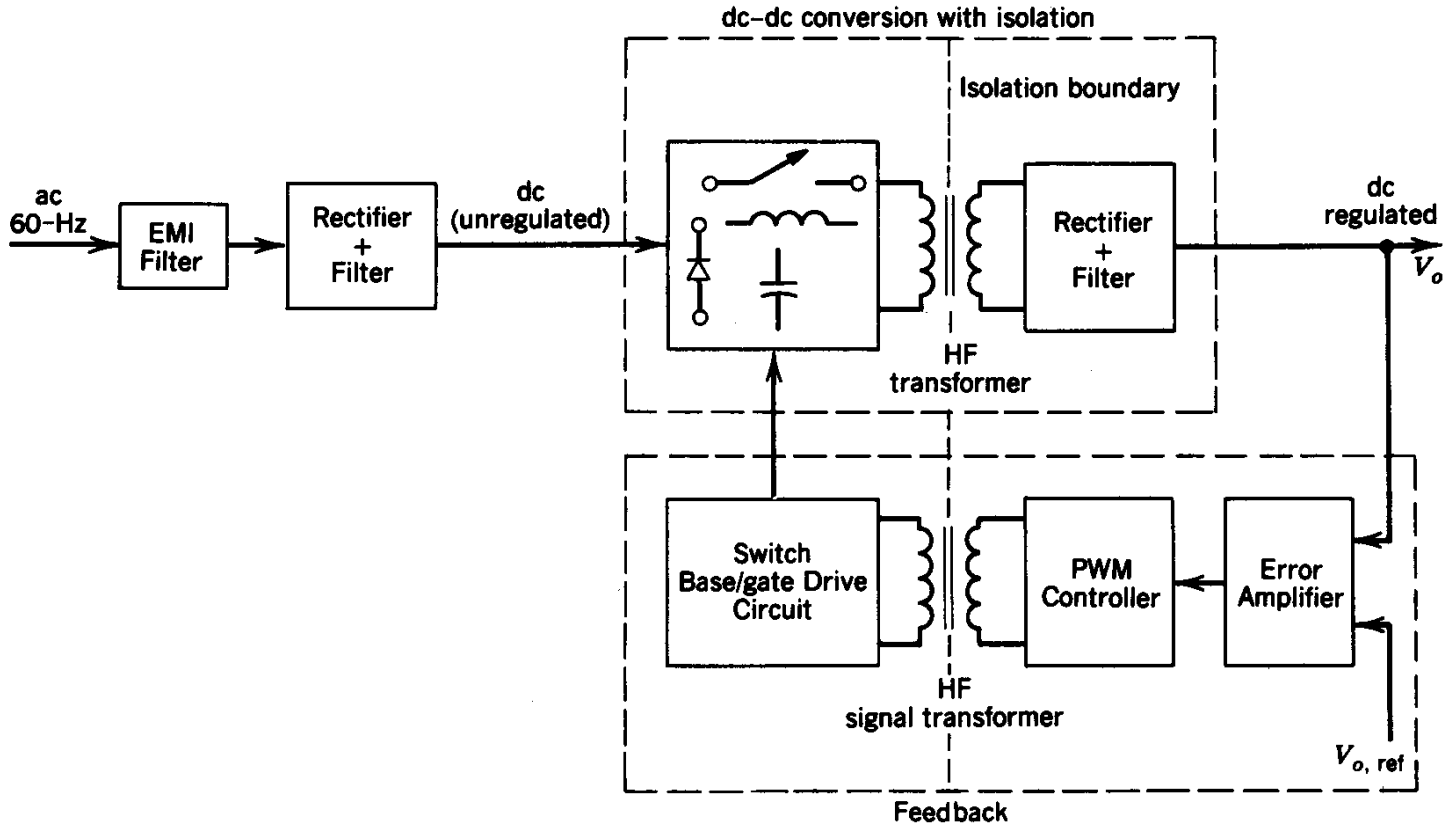
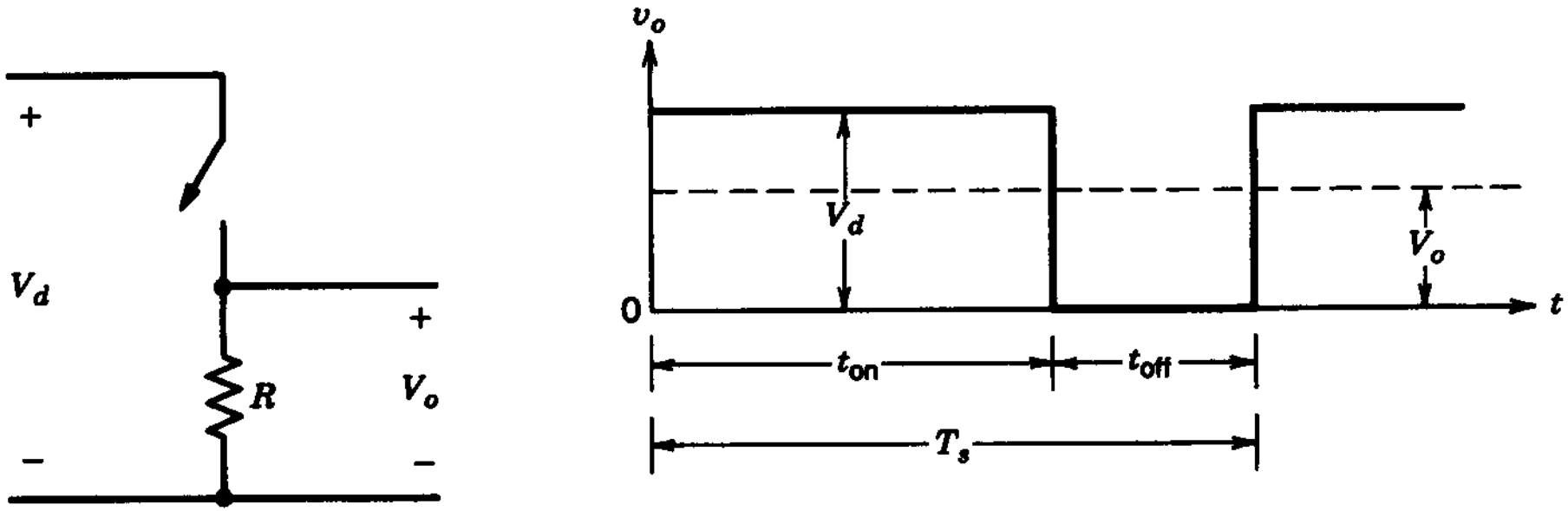


Figure 10-2 Schematic of a switch-mode dc power supply.

- High efficiency and small weight and size

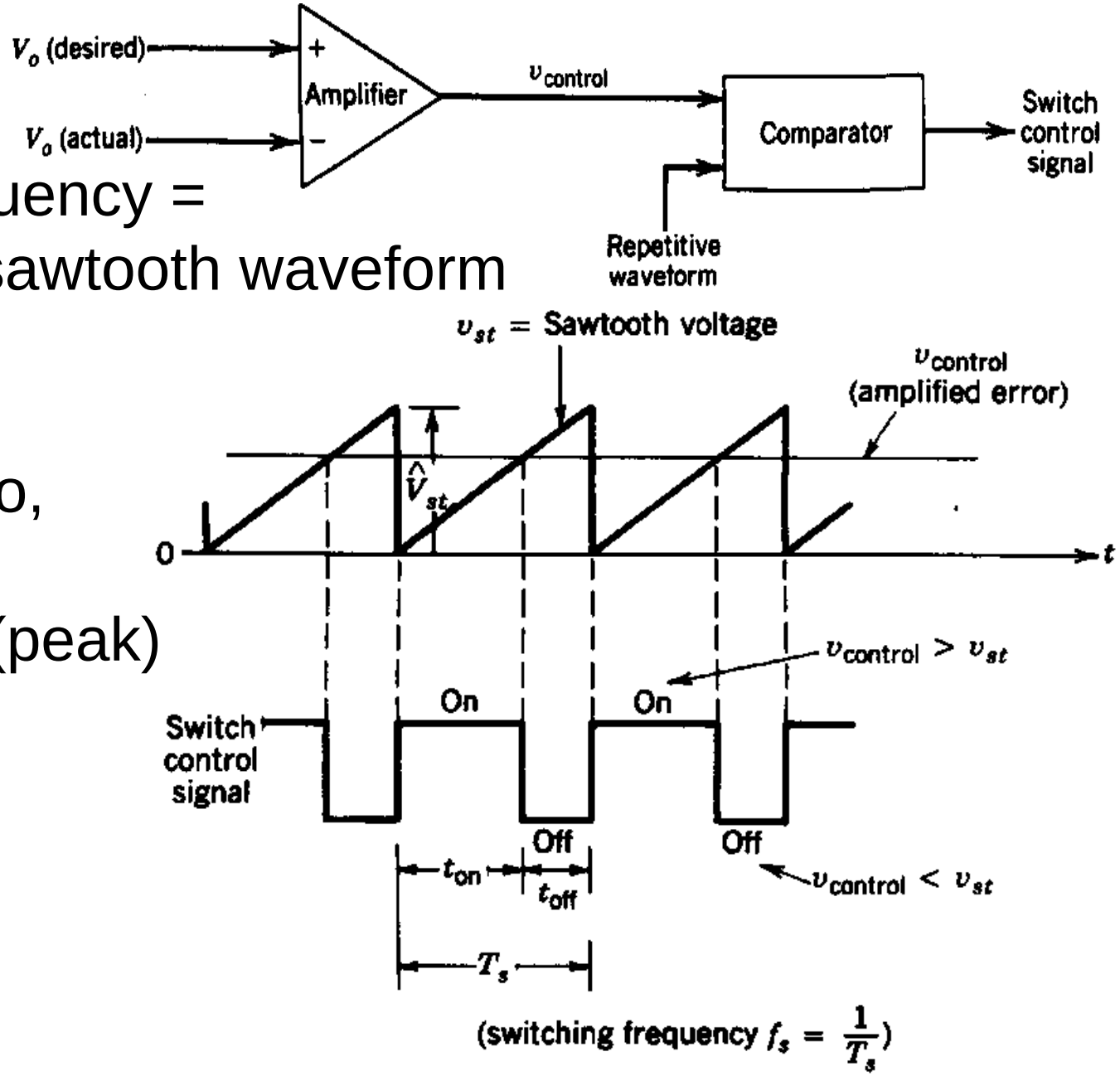
Stepping Down a DC Voltage



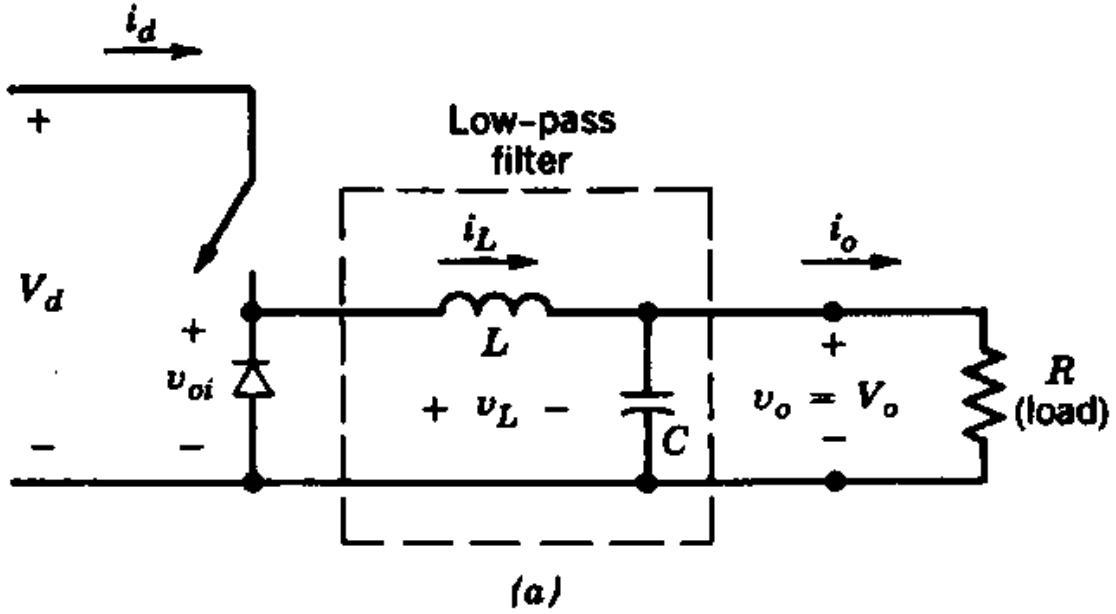
Pulse-Width Modulation in DC-DC Converters

Switching frequency =
frequency of sawtooth waveform

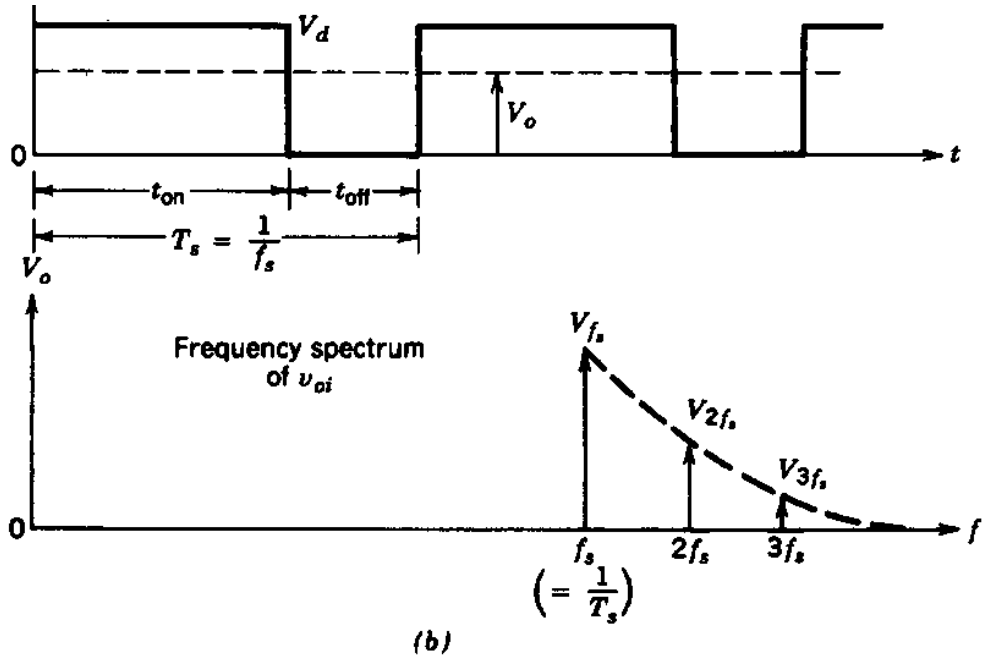
Switch duty ratio,
$D = t_{on} / T_s$
$= v_{control} / V_{st}$ (peak)



Step-Down DC-DC Converter (BUCK Converter)



(a)



(b)

- Pulsating input to the low-pass filter

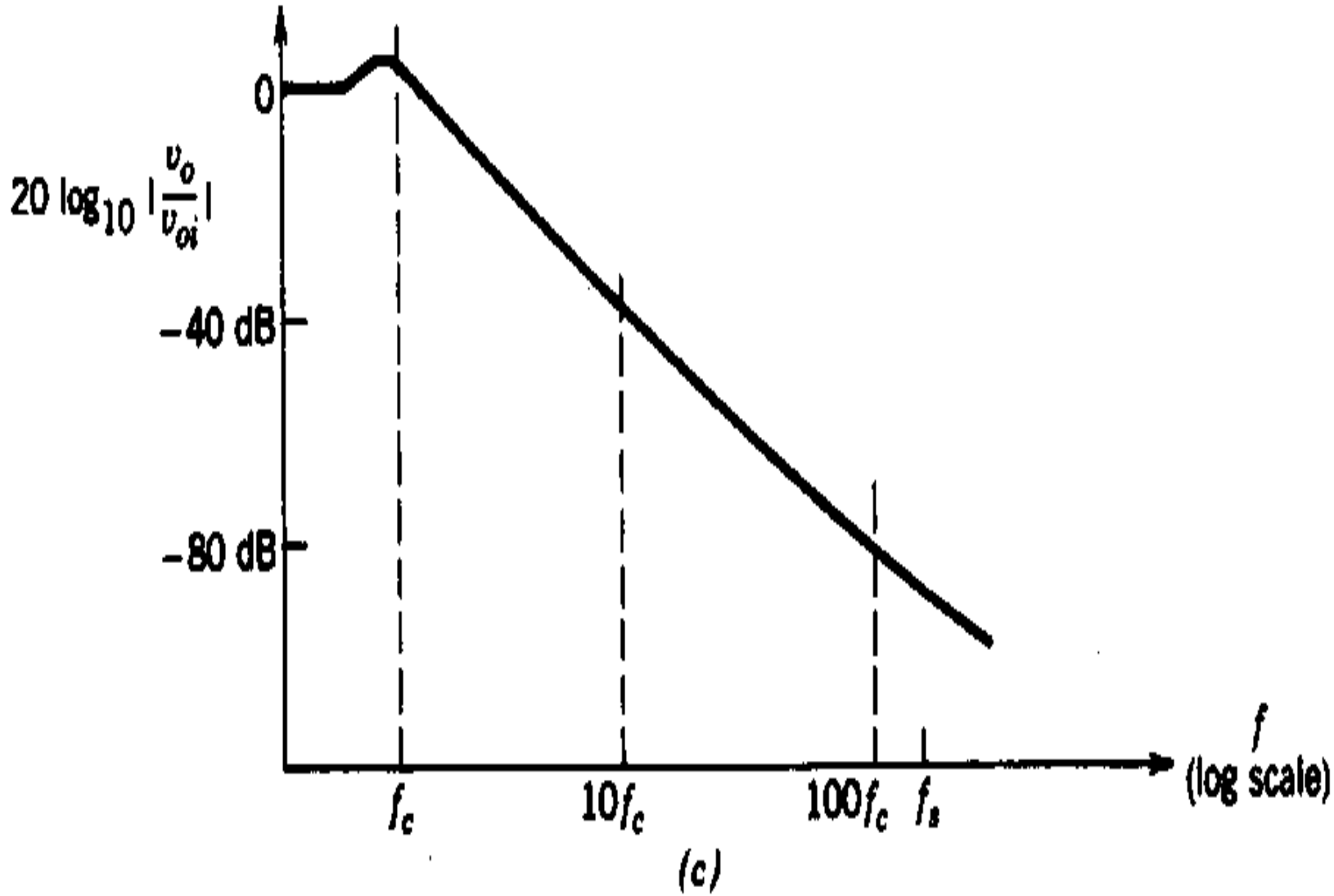


Figure 7-4 Step-down dc–dc converter.

Step-Down DC-DC Converter: Waveforms

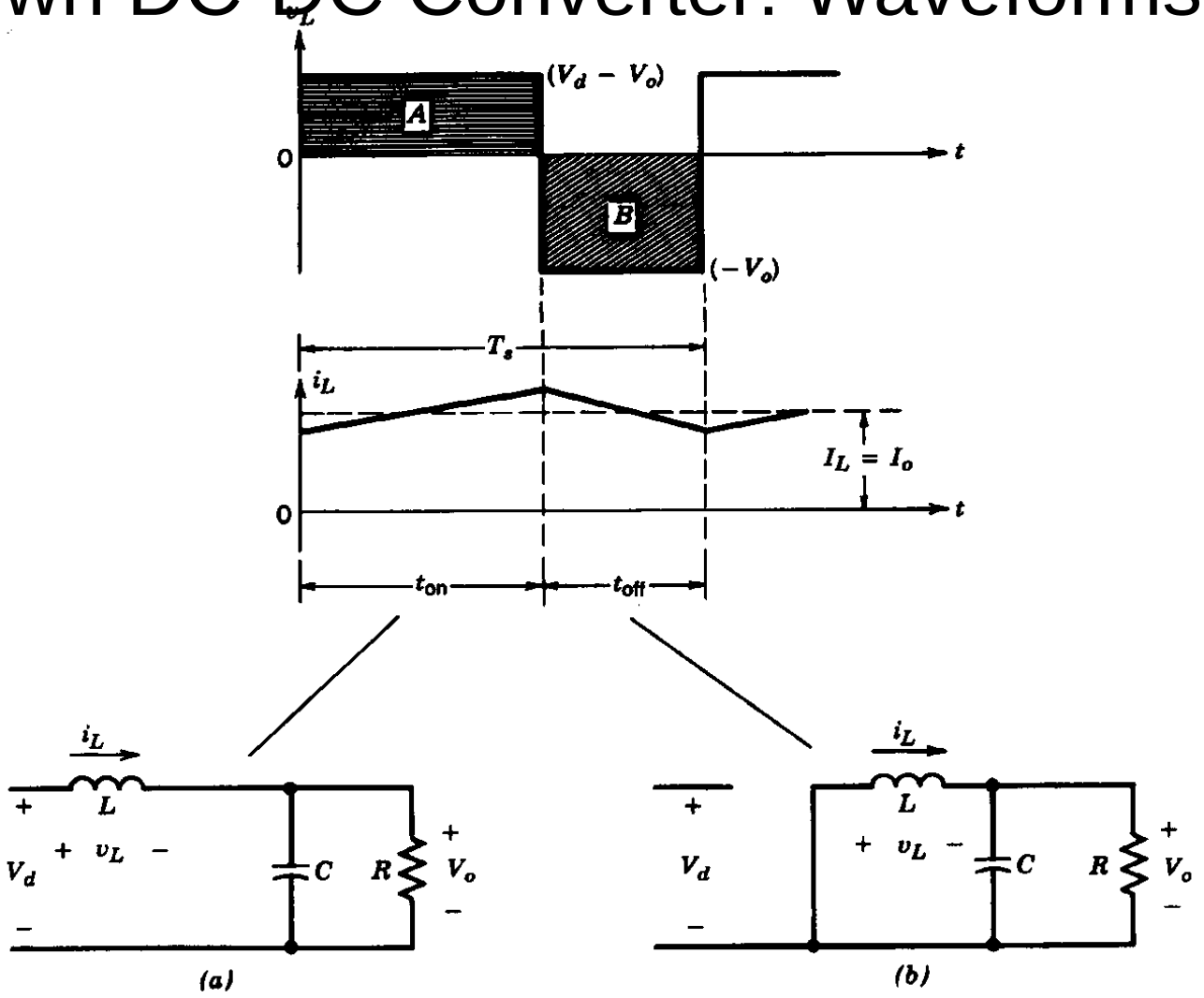


Figure 7-5 Step-down converter circuit states (assuming i_L flows continuously): (*a*) switch on; (*b*) switch off.

$$\int_0^{T_s} v_L \, dt = \int_0^{t_{on}} v_L \, dt + \int_{t_{on}}^{T_s} v_L \, dt = 0$$

In Fig. 7-5, the foregoing equation implies that the areas A and B must be equal. Therefore,

$$(V_d - V_o)t_{on} = V_o(T_s - t_{on})$$

or

$$\frac{V_o}{V_d} = \frac{t_{on}}{T_s} = D \quad \text{(duty ratio)} \tag{7-3}$$

Step-Down Conv.: Output Voltage Ripple

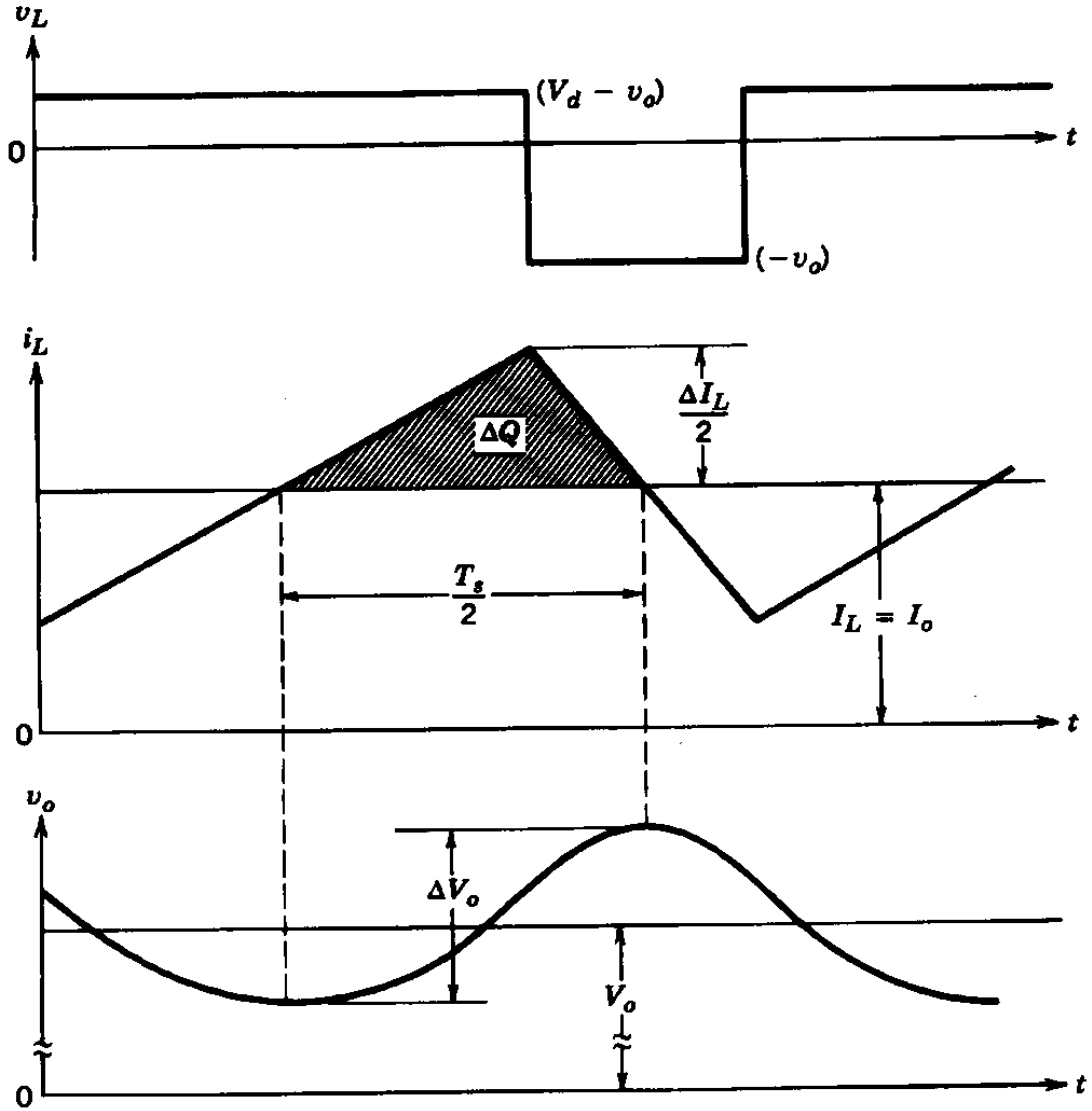


Figure 7-10 Output voltage ripple in a step-down converter.

$$\Delta V_o = \frac{\Delta Q}{C} = \frac{1}{C}\frac{1}{2}\frac{\Delta I_L}{2}\frac{T_s}{2}$$

$$\Delta I_L = \frac{V_o}{L}(1 - D)T_s$$

erefore, substituting ΔI_L

$$\Delta V_o = \frac{T_s}{8C}\frac{V_o}{L}(1 - D)T_s$$

$$\therefore \frac{\Delta V_o}{V_o} = \frac{1}{8}\frac{T_s^2(1 - D)}{LC} = \frac{\pi^2}{2}(1 - D)\left(\frac{f_c}{f_s}\right)^2$$

e switching frequency $f_s = 1/T_s$ and

$$f_c = \frac{1}{2\pi\sqrt{LC}}$$

Step-Up DC-DC Converter (Boost Converter)

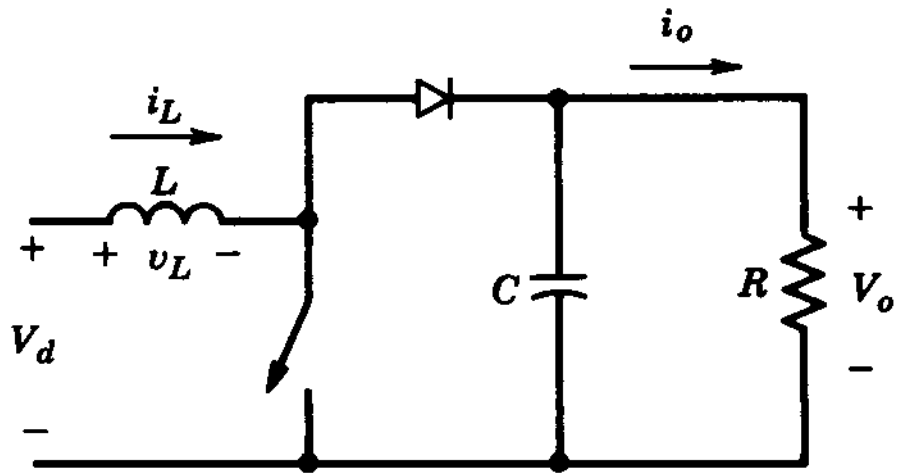


Figure 7-11 Step-up dc–dc converter.

- Output voltage must be greater than the input

Continuous current conduction mode

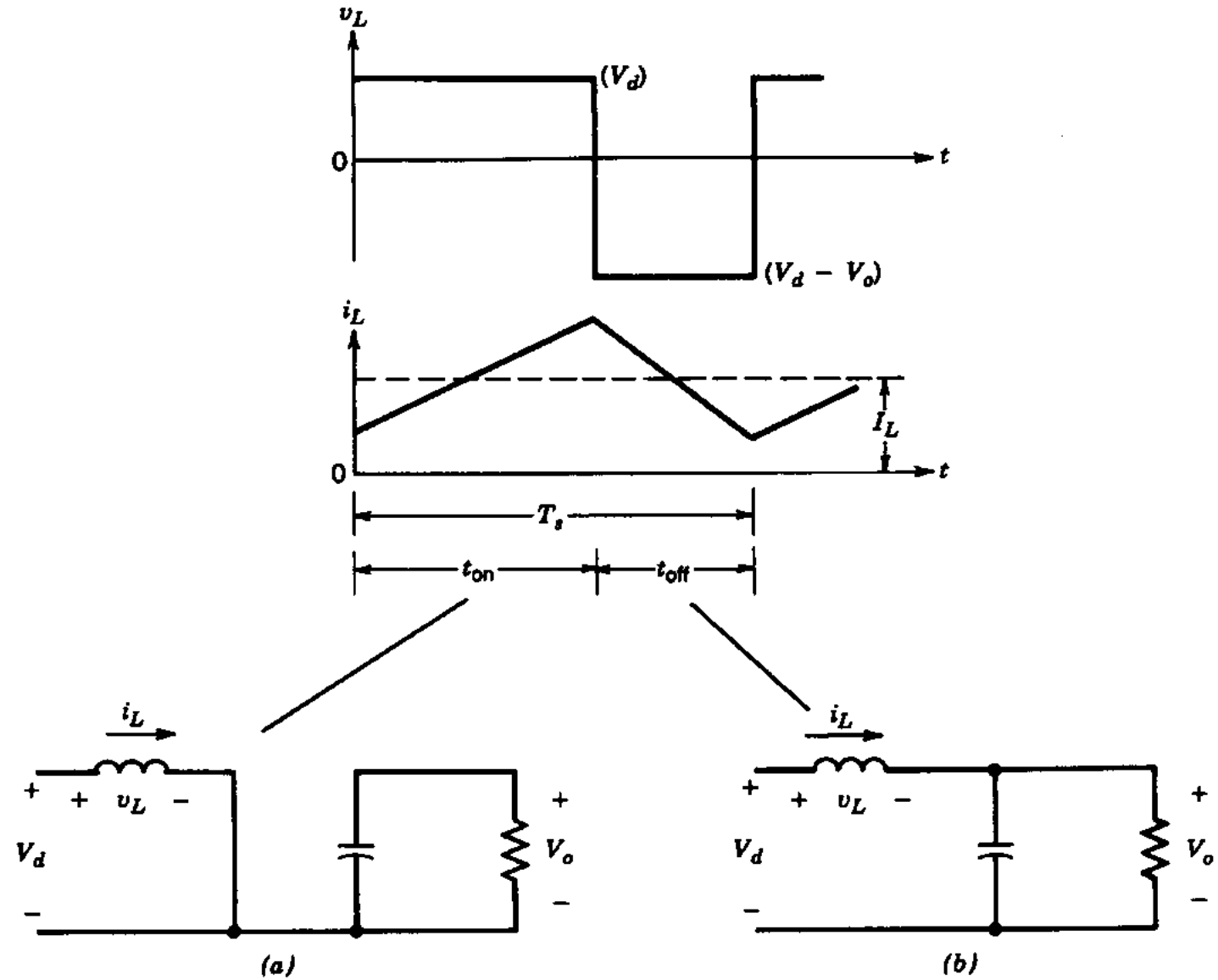


Figure 7-12 Continuous-conduction mode: (*a*) switch on; (*b*) switch off.

7-4-1 CONTINUOUS-CONDUCTION MODE

Figure 7-12 shows the steady-state waveforms for this mode of conduction where the inductor current flows continuously [$i_L(t) > 0$].

Since in steady state the time integral of the inductor voltage over one time period must be zero,

$$V_d t_{on} + (V_d - V_o)t_{off} = 0$$

$$\frac{V_o}{V_d} = \frac{T_s}{t_{off}} = \frac{1}{1 - D}$$

Assuming a lossless circuit, $P_d = P_o$,

$$\therefore V_d I_d = V_o I_o$$

and

$$\frac{I_o}{I_d} = (1 - D)$$

7-4-5 OUTPUT VOLTAGE RIPPLE

The peak-to-peak ripple in the output voltage can be calculated by considering the waveforms shown in Fig. 7-17 for a continuous mode of operation. Assuming that all the ripple current component of the diode current i_D flows through the capacitor and its average value flows through the load resistor, the shaded area in Fig. 7-17 represents charge ΔQ. Therefore, the peak–peak voltage ripple is given by

$$\Delta V_o = \frac{\Delta Q}{C} = \frac{I_o D T_s}{C} \quad \text{(assuming a constant output current)}$$

$$= \frac{V_o}{R}\frac{DT_s}{C} \tag{7-39}$$

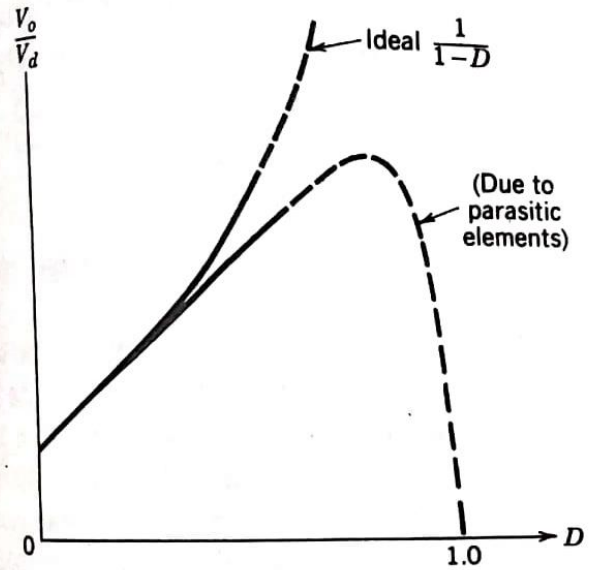


Figure 7-16 Effect of parasitic elements on voltage conversion ratio (step-up converter).

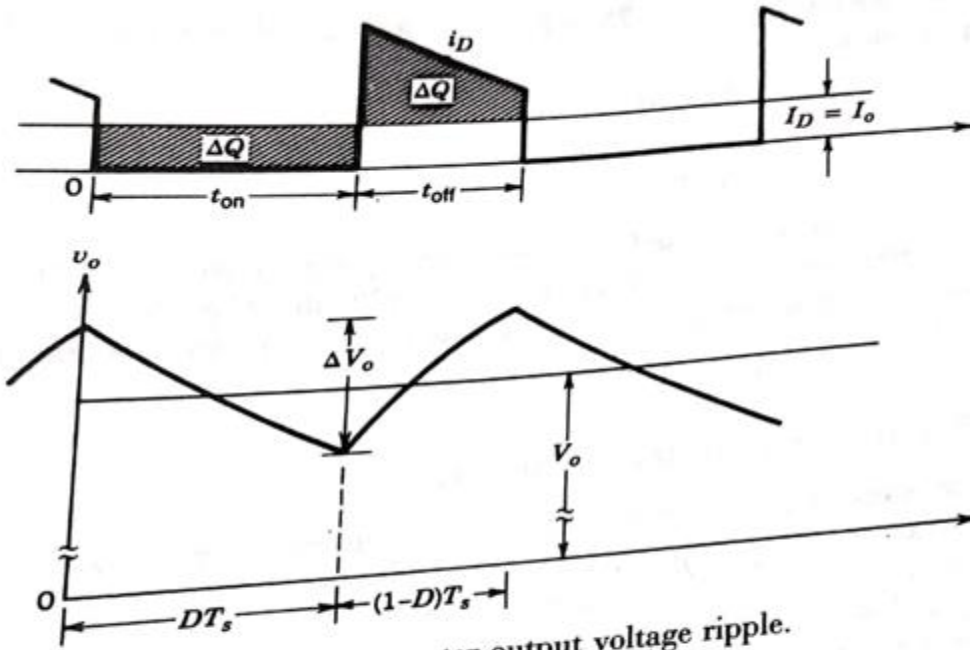


Figure 7-17 Step-up converter output voltage ripple.

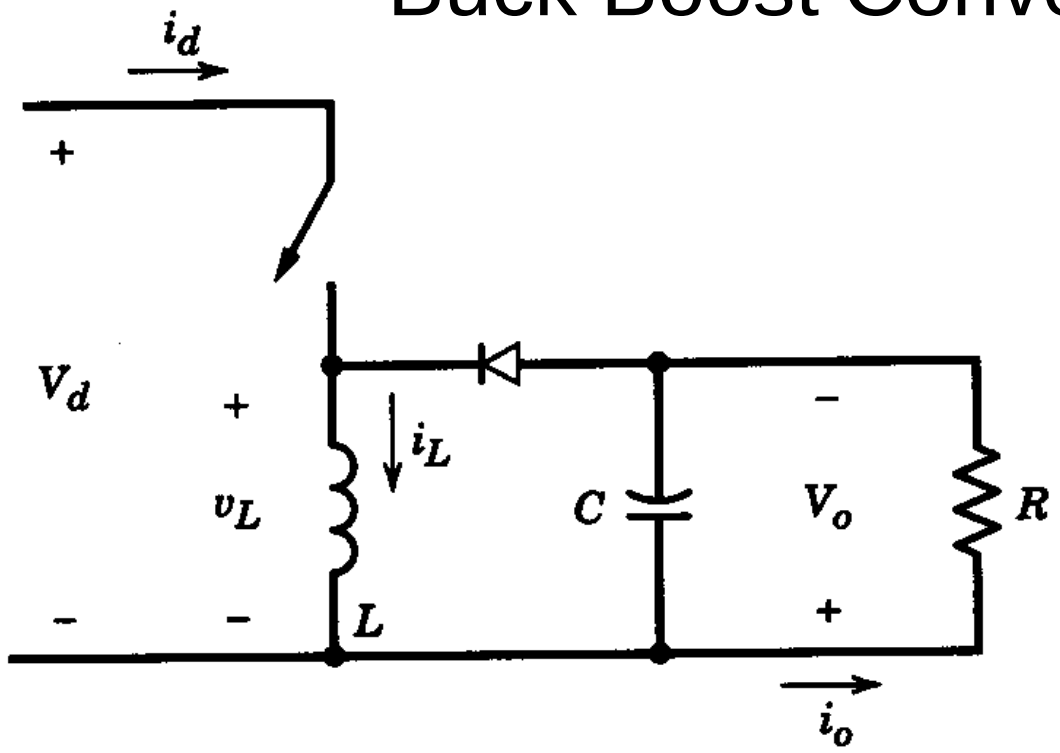
$$\therefore \frac{\Delta V_o}{V_o} = \frac{DT_s}{RC}$$

$$= D\frac{T_s}{\tau} \quad \text{(where } \tau = RC \text{ time constant)}$$

for the discontinuous mode

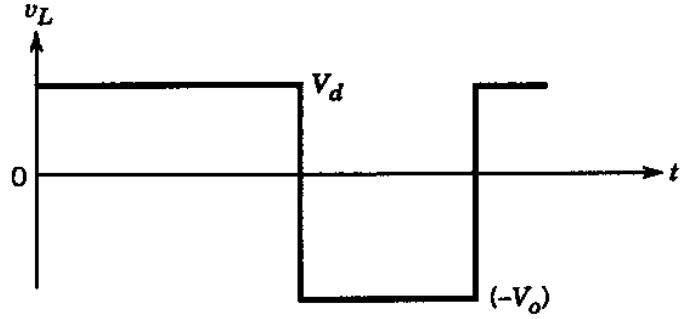
Step-Down/Up DC-DC Converter

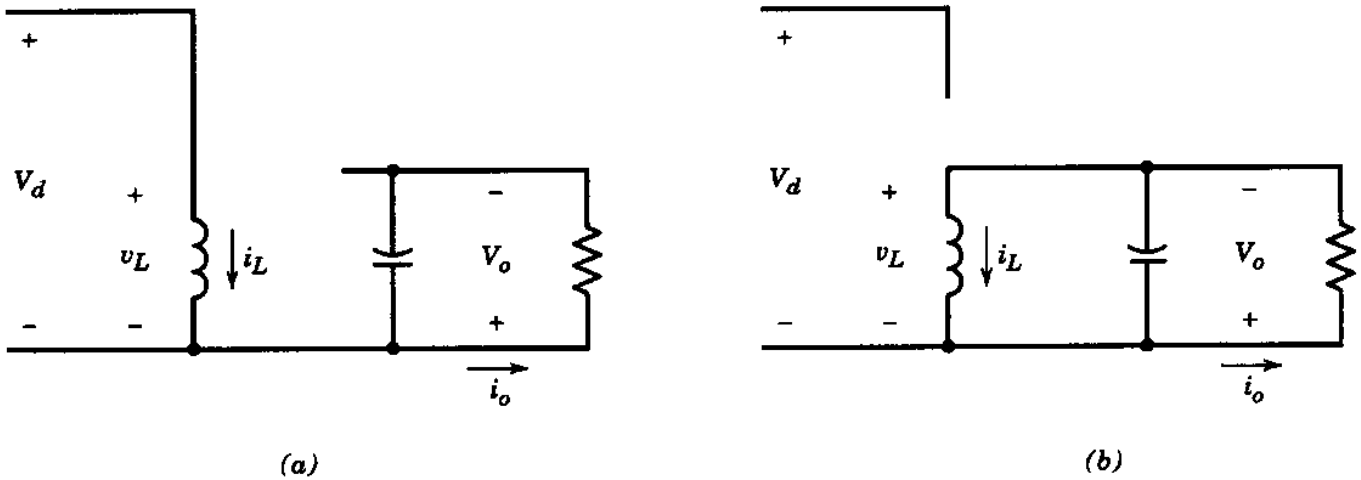
Buck Boost Converter



• The output voltage can be higher or lower than the input voltage

Step-Up DC-DC Converter: Waveforms

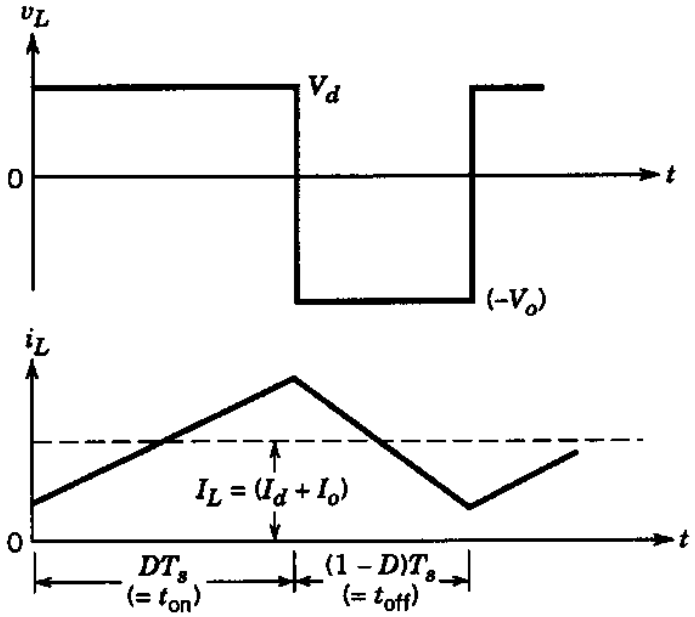




- Continuation conduction mode

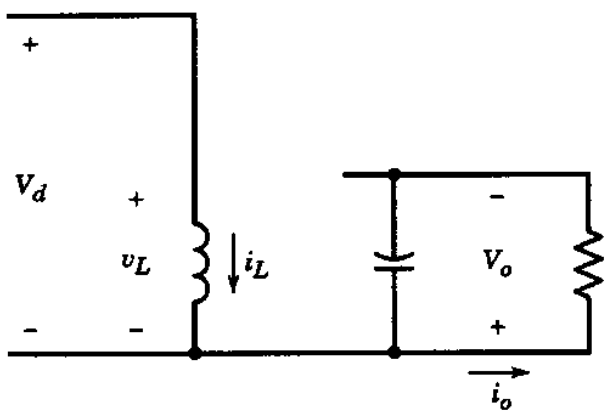
Step-Up DC-DC Converter: Waveforms

- Continuation conduction mode

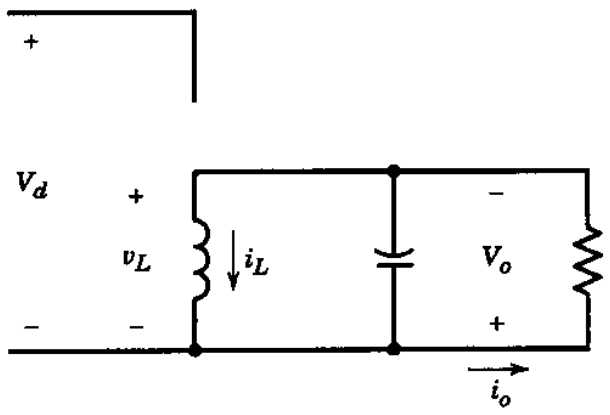


$$V_d DT_s + (-V_o)(1 - D)T_s = 0$$

$$\therefore \frac{V_o}{V_d} = \frac{D}{1 - D}$$



(a)



(b)

7-5-5 OUTPUT VOLTAGE RIPPLE

The ripple in the output voltage can be calculated by considering the waveform show
Fig. 7-24 for a continuous mode of operation. Assuming that all the ripple cur
component of i_D flows through the capacitor and its average value flows throu
resistor, the shaded area in Fig. 7-24 represents charge ΔQ. Therefore, the pe
voltage ripple is calculated as

$$\Delta V_o = \frac{\Delta Q}{C} = \frac{I_o D T_s}{C} \quad \text{(assuming a constant output current)}$$

$$= \frac{V_o}{R}\frac{DT_s}{C}$$

$$\frac{\Delta V_o}{V_o} = \frac{DT_s}{RC}$$

$$= D\frac{T_s}{\tau}$$

